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Chunxi Zhang, Fangfang Xie, Houteng Xi, Shuaiyang Liu, Lijun Yan, Du Xu & Jiayuan Sun

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Robotic-assisted bronchoscopy-guided cryobiopsy for the diagnosis of ground-glass opacity-predominant peripheral pulmonary nodules

Chunxi Zhang^{1,2,3†}, Fangfang Xie^{1,2,3†}, Houteng Xi^{1,2,3}, Shuaiyang Liu^{1,2,3}, Lijun Yan^{1,2,3}, Du Xu^{1,2,3}, Jiayuan Sun^{1,2,3*}

[†]Chunxi Zhang and Fangfang Xie contributed equally to this work.

¹Department of Respiratory Endoscopy, Shanghai Chest Hospital, Shanghai Jiao Tong University School of Medicine, Shanghai, China

²Department of Respiratory and Critical Care Medicine, Shanghai Chest Hospital, Shanghai Jiao Tong University School of Medicine, Shanghai, China

³Shanghai Engineering Research Center of Respiratory Endoscopy, Shanghai, China

*Correspondence: xkyyjysun@163.com (Jiayuan Sun)

Abstract

Background: The combination of shape-sensing robotic-assisted bronchoscopy (ssRAB) and cryobiopsy may improve the diagnostic yield of ground-glass opacities (GGOs). However, the literature on this approach is limited. This study aims to assess the diagnostic yield and safety of ssRAB integrated with mobile cone-beam computed tomography (CBCT)-guided cryobiopsy for GGO-predominant peripheral pulmonary nodules (PPNs).

Methods: This is a single-center retrospective study of patients suspected to have malignant GGO-predominant PPNs (consolidation-to-tumor ratio ≤ 0.5)

who underwent ssRAB integrated with mobile CBCT-guided cryobiopsy between November 2024 and June 2025 at Shanghai Chest Hospital. Clinicodemographic data, nodule characteristics, procedure details, diagnostic performance, and complications were analyzed. The primary efficacy outcome was the diagnostic yield. The primary safety outcome was procedure-related and device-related complications.

Results: A total of 55 nodules were biopsied in 40 patients. The mean patient age was 59.2 years, and 75.0% were female. The median nodule size was 12.0 mm (interquartile range, 9.9–16.0) in the largest dimension. Thirty-six nodules were pure GGOs and 19 were mixed GGOs, with 94.5% of nodules lacking a bronchus sign. A tunnel to the target lesion was created in 74.5% of the nodules. The per-nodule diagnostic yield was 87.3%. There were no episodes of major bleeding or pneumothorax. Cryobiopsy provided specimens sufficient for assessment of programmed death-ligand 1 (PD-L1) expression in 11 (84.6%) of 13 patients tested and for genetic testing in all 16 patients (100%) in which it was attempted.

Conclusions: ssRAB integrated with mobile CBCT-guided cryobiopsy had a high diagnostic yield for GGO-predominant PPNs with a favorable safety profile.

Clinical trial number: Not applicable.

Keywords: ground-glass opacity, diagnosis, robotic-assisted bronchoscopy, cone-beam computed tomography, cryobiopsy

Background

Lung cancer is the leading cause of cancer-related morbidity and mortality worldwide [1]. With the widespread adoption of chest computed tomography (CT) in lung cancer screening, the detection rate of pulmonary ground-glass opacities (GGOs) has been increasing. While GGOs can represent early-stage lung cancer, which often has a favorable prognosis, they may also be caused by benign conditions, such as inflammation or focal fibrosis [2]. Distinguishing between benign and malignant GGOs remains challenging. For GGOs suspected of malignancy, non-surgical biopsy plays an important role in confirming the pathological diagnosis, especially in patients who are medically inoperable or at high surgical risk. For high-risk patients, confirming the diagnosis through non-surgical biopsy can help avoid unnecessary surgical trauma [3]. For those who are medically inoperable and instead opt for alternative treatments such as ablation and radiotherapy, non-surgical biopsy provides essential pathological support [3,4].

Transbronchial biopsy is a valuable non-surgical biopsy method that is generally associated with a more favorable safety profile compared to transthoracic biopsy [5,6]. However, due to factors such as small lesion size, difficult-to-reach locations, lack of a bronchus sign, challenges in intraoperative localization, and difficulties in obtaining adequate specimens, transbronchial biopsy of GGOs is challenging. Shape-sensing robotic-assisted bronchoscopy (ssRAB) is a novel bronchoscopy technique [7]. Compared to traditional guided bronchoscopy techniques such as virtual bronchoscopic

navigation (VBN), electromagnetic navigational bronchoscopy (ENB), and radial endobronchial ultrasound (rEBUS), the improved flexibility, stability, and precision of ssRAB increase accessibility to the target lesion [8,9]. Furthermore, the IonTM ssRAB system (Intuitive Surgical, Sunnyvale, CA, USA) now allows direct integration with mobile cone-beam computed tomography (CBCT; Cios Spin, Siemens Healthineers, Erlangen, Germany), providing direct access and processing of mobile CBCT data for target update, which can overcome CT-to-body divergence and further facilitate accurate transbronchial diagnosis of GGOs [10,11].

Cryobiopsy is an efficient sampling technique. Compared to conventional sampling techniques such as forceps and needle, it offers advantages such as an ability to obtain larger samples and a preserved tissue pattern [12-14], and has been increasingly used in recent years [15-17]. Previous studies have demonstrated that performing cryobiopsy with a 1.1-mm cryoprobe following conventional biopsy is both feasible and safe for diagnosis of peripheral pulmonary lesions, with a diagnostic yield that is 18.2%–20% better than that of conventional biopsy alone [18,19]. For GGOs, we have previously reported favorable diagnostic yields of 71.9%–82.6% using cryobiopsy with traditional guided bronchoscopy [14,20].

Given the respective advantages of each technique, the combination of ssRAB and cryobiopsy may improve the diagnostic yield for GGOs. However, there is a lack of literature focusing on this approach. In this study, we aimed to assess the diagnostic yield and safety of ssRAB integrated with mobile

CBCT-guided cryobiopsy for GGO-predominant peripheral pulmonary nodules (PPNs).

Methods

Study design

This is a single-center retrospective study evaluating the diagnostic yield and safety of ssRAB integrated with mobile CBCT-guided cryobiopsy for GGO-predominant PPNs. Consecutive patients suspected to have malignant GGO-predominant PPNs who underwent ssRAB integrated with mobile CBCT-guided cryobiopsy between November 2024 and June 2025 at Shanghai Chest Hospital were included. Patients who met the following criteria underwent the study procedure at our center. The inclusion criteria were: (1) age ≥ 18 years; (2) thin-slice chest CT showing GGO-predominant PPNs (diameter 8–30 mm, consolidation-to-tumor ratio ≤ 0.5); (3) patients who were unsuitable for or declined surgery, or who requested a non-surgical biopsy to confirm the diagnosis before surgery; (4) good compliance and provision of signed informed consent; and at least one of the following clinical indications: (5) mixed GGOs with a solid component > 8 mm or persistent mixed GGOs on follow-up CT, with clinical suspicion of malignancy and deemed to require biopsy [3,21]; (6) pure GGOs > 10 mm with persistence on follow-up CT, or pure GGOs showing growth or an increase in solid component on follow-up CT, with clinical suspicion of malignancy and deemed to require biopsy [3,21]; (7) pure GGOs ≤ 10 mm that remained stable on follow-up CT, but with clinical

suspicion of malignancy and significant patient anxiety leading to a request for biopsy. The exclusion criteria were: (1) contraindications to bronchoscopy or general anesthesia; (2) presence of concomitant endobronchial lesions observed during bronchoscopy; and (3) other conditions that the investigators considered not suitable for the subjects to undergo the study procedure. Patient and nodule characteristics, procedure details, diagnostic performance, and complications were obtained and analyzed. This study was approved by the Ethics Committee of Shanghai Chest Hospital (IS25249) and the requirement for informed consent was waived. The study was performed in accordance with the Declaration of Helsinki.

Study procedure

Pre-procedure thin-slice chest CT scans were uploaded into the PlanPoint Planning Laptop (Intuitive Surgical) for planning of navigational pathways to the target nodule. All procedures were performed under general anesthesia with endotracheal intubation and mechanical ventilation by two experienced robotic bronchoscopists (J.S. and F.X., each with experience of >100 ssRAB procedures). First, a conventional bronchoscope was used to rule out endobronchial abnormalities and to clear secretions. Next, the ssRAB system (Ion) was docked to the patient's endotracheal tube, after which the peripheral vision probe was inserted into the working channel of the catheter. Then, the catheter (with a working channel of 2 mm) was inserted into the patient's airway through the endotracheal tube. The catheter was then driven using the controller to complete registration and navigated towards the

target nodule. After that, mobile CBCT (Cios Spin) was performed with transmission of the images to the ssRAB system for target update. If necessary, adjustments of the catheter were made according to the updated nodule position. A 1.1-mm cryoprobe (Erbe, Tübingen, Germany) was then advanced into the nodule through an ideal bronchial pathway.

For nodules with no ideal bronchial access, a tunnel to the target nodule was established using a Flexision 21G biopsy needle (Intuitive Surgical) and dilated using the needle sheath as we previously reported [22]. Specifically, the robotic catheter was navigated to the bronchus in close proximity to the target nodule. After target update and adjustment of the catheter, needle puncture was performed according to the updated nodule position. After mobile CBCT scanning confirmed that the needle tip was within the target nodule, a tunnel to the target nodule was established by needle puncture and dilated using the needle sheath. This tunnel extended through the peripheral bronchial wall and lung parenchyma, reaching the interior of the target nodule. The 1.1-mm cryoprobe was then deployed to the target nodule via the established tunnel. The primary biopsy tool was the 1.1-mm cryoprobe. The needle was used primarily to establish the tunnel and was not intended as a biopsy tool. In addition, to minimize the risk of bleeding that might interfere with subsequent cryobiopsy, the number of needle passes was deliberately limited. Typically, the needle was advanced only about three times per nodule, which limited its ability to obtain specimens.

After mobile CBCT confirming that the cryoprobe was within the lesion,

cryobiopsy was performed. The number of cryobiopsy specimens obtained and the freeze duration were determined by the bronchoscopist. Typically, 2–3 samples were obtained, with a freeze duration of 3–4 seconds per sample. After each cryobiopsy, the cryoprobe with the specimen attached was withdrawn through the working channel of the catheter while leaving the catheter in place. This allowed the catheter to provide mechanical hemostatic tamponade at the biopsy site and facilitated repeat biopsies without the need for renavigation [10,11]. No other biopsy tool (e.g., forceps or a brush) was used in this study. Rapid on-site evaluation was not performed. After cryobiopsy, the vision probe was re-inserted to inspect for airway bleeding or other complications. A final mobile CBCT scan was performed on completion of the procedure to check for pneumothorax and bleeding. Figure 1 shows a representative case. All patients were observed overnight.

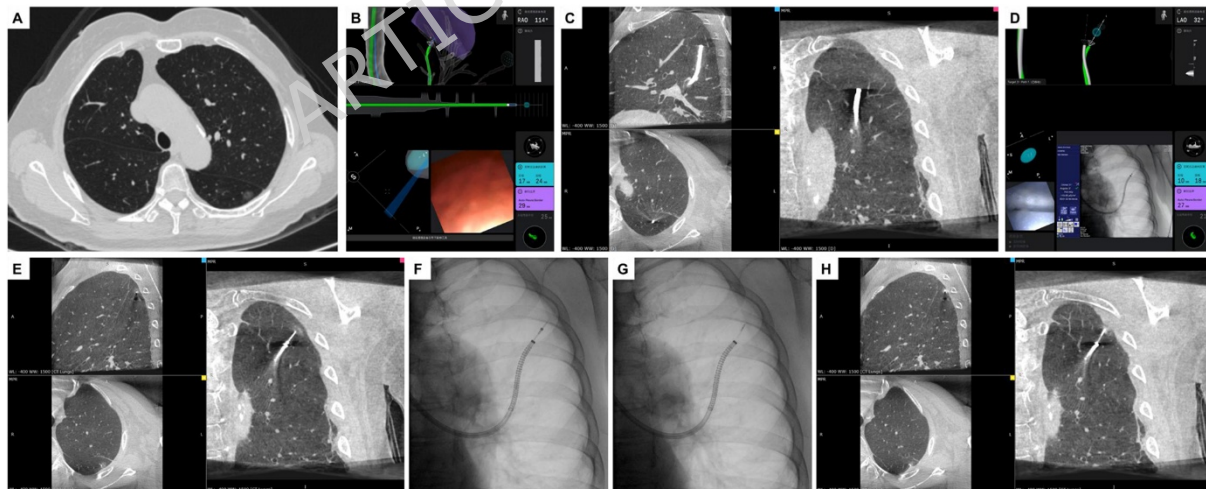


Figure 1. Images for a representative case. A) Pre-procedure chest CT scan showing a pure GGO in the left lower lobe. B) The catheter was navigated to the target bronchus. C) A mobile CBCT scan obtained after navigation. D) The mobile CBCT

data were transmitted to the ssRAB system for target update. Needle puncture was performed according to the updated nodule position. E) Confirmation of tool (needle)-in-lesion by mobile CBCT. F) A tunnel was created by needle puncture and dilated using the needle sheath. G) The cryoprobe was deployed. H) Cryobiopsy was performed after tool (cryoprobe)-in-lesion confirmation by mobile CBCT. The nodule was successfully diagnosed as pulmonary adenocarcinoma. CBCT, cone-beam computed tomography; GGO, ground-glass opacity; ssRAB, shape-sensing robotic-assisted bronchoscopy.

Immunohistochemistry and genetic testing

All cryobiopsy specimens were placed in formalin and submitted for pathological examination. All diagnostic findings were confirmed by immunohistochemistry. In malignant cases, specimens underwent immunohistochemical testing for programmed death-ligand 1 (PD-L1) expression and were sent out for genetic testing based on clinical necessity as determined by the physician and with patient consent.

Study outcomes

The primary efficacy outcome was the diagnostic yield, which was defined as the number of diagnostic nodules divided by the total number of nodules biopsied. Diagnostic yield was calculated using the strict definition recommended by the ATS/ACCP consensus statement (i.e., a procedure was deemed diagnostic if a specific benign or malignant diagnosis was made), with one modification: atypical adenomatous hyperplasia (AAH) was also considered a diagnostic result [23,24]. A diagnostic biopsy yielding a

premalignancy or malignancy was considered a true positive unless a different diagnosis was subsequently confirmed by surgery, that is, a false positive. A true negative was defined as a nonmalignant biopsy in which either subsequent diagnostic testing (e.g. CT-guided biopsy or surgery) confirmed a nonmalignant diagnosis or follow-up imaging demonstrated a decrease in size or complete resolution of the lesion. A false negative was defined as an initially nonmalignant biopsy in which follow-up diagnostic testing revealed malignancy, repeat imaging showed lesion growth, or presumptive treatment based on multidisciplinary discussion was initiated without a confirmed diagnosis. Diagnostic accuracy was calculated as the sum of true positives and true negatives divided by the total number of nodules biopsied. All patients were followed up for at least six months postoperatively. The primary safety outcome was the frequency of procedure-related and device-related complications, including (1) pneumothorax, (2) major bleeding defined as Grade 2 or higher according to the Nashville Bleeding Scale [25], and (3) any other complications related to the procedure or device. The safety outcome was investigated for up to one month postoperatively.

Statistical analysis

Continuous variables are shown as the mean $\pm$ standard deviation or median [interquartile range (IQR)] and were compared between groups using the *t*-test or Mann-Whitney *U* test. The Shapiro-Wilk test was used to test for normality. Categorical variables are summarized as the number (percentage)

and were compared between groups using Pearson's chi-squared test or Fisher's exact test. A p-value of <0.05 was considered statistically significant. All statistical analyses were performed using SPSS statistical software version 24.0 (IBM Corp., Armonk, NY, USA).

Results

Patient and nodule characteristics

Fifty-five nodules were biopsied in the 40 patients who met the eligibility criteria. Thirty-six (65.5%) of these nodules were pure GGOs and 19 (34.5%) were mixed GGOs. The median nodule size in the largest dimension was 12.0 mm (IQR, 9.9–16.0). Of the 55 nodules, 94.5% had no bronchus sign. The patient and nodule characteristics are shown in Tables 1 and 2.

Table 1. Patient Characteristics

Characteristic	Value
Age, y	59.2 ± 11.9
Gender	
Female	30 (75.0%)
Male	10 (25.0%)
Personal history of lung cancer	
Yes	13 (32.5%)
No	27 (67.5%)
Personal history of pulmonary surgery	
Yes	17 (42.5%)
No	23 (57.5%)

The data are presented as the mean ± standard deviation or number (percentage) as appropriate.

Table 2. Nodule Characteristics

Characteristic	Value
Nodule size, mm	
Axial	10.3 (8.9-15.3)

Coronal	10.4 (9.0-13.3)
Sagittal	10.8 (8.9-14.2)
Largest dimension	12.0 (9.9-16.0)
Nodule location	
LUL	16 (29.1%)
LLL	8 (14.5%)
RUL	21 (38.2%)
RML	3 (5.5%)
RLL	7 (12.7%)
Nodule appearance	
Pure GGO	36 (65.5%)
Mixed GGO	19 (34.5%)
Distance to pleura/fissure, mm	8.2 (0.0-20.8)
Bronchus sign	
Present	3 (5.5%)
Absent	52 (94.5%)

The data are presented as the median (IQR) or number (percentage) as appropriate.

GGO, ground-glass opacity; IQR, interquartile range; LLL, left lower lobe; LUL, left upper lobe; RLL, right lower lobe; RML, right middle lobe; RUL, right upper lobe.

Procedure characteristics

A tunnel to the target lesion was established in 74.5% (41/55) of the nodules. The median number of cryobiopsy specimens obtained per nodule was 3 (IQR, 2-3), and the median freeze duration was 3 seconds (IQR, 3-4). The median procedure time was 64.5 minutes (IQR, 45.0-85.8).

Diagnostic yield

As shown in Table 3, a specific diagnosis was made for 48 nodules (in 35 patients) based on cryobiopsy specimens, indicating a per-nodule diagnostic yield of 87.3% (48/55) and a per-patient diagnostic yield of 87.5% (35/40). The diagnostic accuracy was 87.3% (48/55) and the false negative rate was 12.7% (7/55). The prevalence of malignancy was 90.9% (50/55). There was no significant difference in the diagnostic yield according to nodule

characteristics (Table 4).

The 35 patients in whom a specific diagnosis was made were treated as follows: seven with pulmonary adenocarcinoma underwent surgical resection, with postoperative pathology confirming adenocarcinoma in all cases; 21 with pulmonary adenocarcinoma underwent transbronchial or percutaneous ablation; one with pulmonary adenocarcinoma was treated using a combined approach of surgery and transbronchial ablation for two lesions; one with pulmonary adenocarcinoma received targeted drug therapy; one with pulmonary adenocarcinoma was recommended for surgery but declined and chose to pursue traditional Chinese medicine; one with lymphoma was referred to the hematology department for specialized management; and three with AAH were initially recommended for CT surveillance, but in view of significant patient anxiety regarding the potential for progression and after multidisciplinary discussion of the risks and benefits, these patients opted for ablation therapy.

Table 3. Diagnoses Based on Cryobiopsy Specimens

Diagnoses	No. (%)
Diagnostic	48 (87.3)
Atypical adenomatous hyperplasia	5 (9.1)
Pulmonary adenocarcinoma	41 (74.5)
Adenocarcinoma with lepidic pattern	21 (38.2)
Adenocarcinoma with lepidic and acinar patterns	14 (25.5)
Adenocarcinoma with acinar pattern	3 (5.5)

Adenocarcinoma with acinar and papillary patterns	1 (1.8)
Adenocarcinoma with acinar and micropapillary patterns	1 (1.8)
Adenocarcinoma with acinar and complex glandular patterns	1 (1.8)
Lymphoma	2 (3.6)
Nondiagnostic	7 (12.7)
Non-specific inflammation*	4 (7.3)
Normal pulmonary elements**	3 (5.5)

*One of these four nodules was diagnosed as pulmonary adenocarcinoma by surgery, and the remaining three were considered malignant by multidisciplinary discussion and subsequently underwent ablation. ** One of these three nodules was diagnosed as pulmonary adenocarcinoma by surgery, and the remaining two were diagnosed as pulmonary adenocarcinoma by repeat ssRAB-guided cryobiopsy.

Table 4. Diagnostic Yield by Nodule Characteristics

Characteristic	Diagnostic yield	<i>P</i>
Overall	48/55 (87.3%)	
Nodule size*		
≤10 mm	12/15 (80.0%)	0.591
>10 mm	36/40 (90.0%)	
Nodule location		
Lower lobe	15/15 (100.0%)	0.201
Non-lower lobe	33/40 (82.5%)	
Nodule appearance		
Pure GGO	30/36 (83.3%)	0.435
Mixed GGO	18/19 (94.7%)	
Lung centrality		
Inner two-thirds	30/35 (85.7%)	0.970
Outer one-third	18/20 (90.0%)	
Bronchus sign		
Absent	45/52 (86.5%)	1.000
Present	3/3 (100.0%)	

*Measured in the largest dimension. GGO, ground-glass opacity.

Immunohistochemistry and genetic testing

PD-L1 expression was evaluated in 13 patients, with only two cases being nonevaluable. Genetic testing was attempted in 16 cases, all of which had cryobiopsy specimens that were sufficient for analysis.

Safety

There were no major bleeding episodes. No pneumothorax or other procedure-related or device-related complications were observed.

Discussion

This study retrospectively assessed 55 nodules from 40 patients to evaluate the diagnostic yield and safety of ssRAB integrated with mobile CBCT-guided cryobiopsy for GGO-predominant PPNs. The study procedure had a diagnostic yield of 87.3%, with no procedure-related or device-related complications. Moreover, cryobiopsy provided specimens sufficient for assessment of PD-L1 expression in 11 of the 13 patients (84.6%) tested and for genetic testing in all 16 patients (100%) in which it was attempted. We acknowledge that, according to the ATS/ACCP consensus definition of strict diagnostic yield, AAH may be classified as nondiagnostic. However, AAH is a premalignant lesion, and in our cohort, all patients diagnosed with AAH underwent ablation after multidisciplinary discussion and providing written informed consent, indicating that the biopsy result was sufficient to guide clinical management. Thus, we considered AAH as a diagnostic finding in this study. To our knowledge, no previous study has specifically investigated the diagnostic yield of ssRAB-guided cryobiopsy for GGO-predominant PPNs.

Traditional guided bronchoscopy, such as VBN, ENB and rEBUS, has been reported to have an overall diagnostic yield of 70% for peripheral pulmonary lesions [5,26]. For the diagnosis of GGO-predominant lesions, Ikezawa et al. used rEBUS with a guide sheath (with or without the assistance of VBN) and reported a diagnostic yield ranging from 57% to 69%, which is suboptimal [27,28]. Furthermore, for lesions <15 mm, their diagnostic yield was only 38%–40% [27,28].

ssRAB is a novel guided bronchoscopy technique that has been widely used due to its enhanced stability, flexibility, and precision and has contributed to a high diagnostic yield of 72%–93.3% for peripheral pulmonary lesions [29–34]. However, there is little information on the diagnostic yield of ssRAB for GGO-predominant lesions. Abia-Trujillo et al. retrospectively assessed the diagnostic yield of ssRAB for pure GGOs and mixed GGOs with a solid component less than 6 mm and reported a commendable diagnostic yield of 87.0% [35]. The main sampling tools used in that study were needles.

Transbronchial biopsy usually uses a forceps, needle, and brush to obtain samples. Compared to these sampling techniques, cryobiopsy provides advantages including larger sample size, preserved tissue pattern, and 360-degree tissue acquisition [12–14]. In our previous experience with traditional guided bronchoscopy, cryobiopsy achieved diagnostic yields of 71.9%–82.6% for GGOs [14,20], outperforming forceps biopsy (57%–69%) [27,28]. Specifically, cryobiopsy performed under the guidance of VBN (or ENB) and fluoroscopy was reported to have a diagnostic yield of 82.6% for GGOs [20].

The combination of cryobiopsy and conventional biopsy (including forceps biopsy and brushing) under the guidance of VBN (or ENB) and CBCT had a diagnostic yield of 71.9% for GGOs [14]. The lower diagnostic yield in the latter study may be primarily the result of use of forceps biopsy as the initial sampling method in 28.1% of the lesions, given that the study found that the diagnostic yield was significantly higher for a cryobiopsy-first sequence than for a forceps biopsy-first sequence (73.9% vs. 43.5%, $p=0.036$) [14].

The diagnostic yield in the present study was 87.3%, which is higher than or comparable to the previously reported diagnostic yield of ssRAB for lesions of all types and sizes, not specifically for GGO-predominant PPNs [7,36]. In the study by Abia-Trujillo et al., cryobiopsy was only used in 9 of 23 lesions [35]. Although their diagnostic yield (87.0%) was similar to that in this study (87.3%), this study included a higher proportion of pure GGOs (65.5% vs. 47.8%), more lesions without a bronchus sign (94.5% vs. 26.1%), and smaller lesions (12.0 mm vs. 18.0 mm). It is noteworthy that this study achieved a high diagnostic yield of 86.5% for lesions without a bronchus sign. In comparison, even when ssRAB was used for solid-predominant lesions, previous studies have reported a diagnostic yield of only 71.1%–72.7% for those without a bronchus sign [30,33]. The high diagnostic yield observed in our study may be attributed to two key factors: first, integrated mobile CBCT was utilized; and second, cryobiopsy was used as the primary biopsy tool.

Transbronchial biopsy is generally considered safer than transthoracic biopsy [5,6]. In this study, no major bleeding, pneumothorax, or other

procedure-related or device-related complications were observed. The complication rate in our study is consistent with that previously reported for ssRAB-guided conventional biopsy (e.g. forceps, needle, or brushing) for solid-predominant peripheral pulmonary lesions [29-34].

In this study, a minimum of three to four mobile CBCT scans were performed per patient. We acknowledge that repeat mobile CBCT scans increase radiation exposure. However, we believe that the first two to three scans were essential to ensure procedural accuracy and safety when performing biopsies of GGO-predominant PPNs. The post-navigation scan allowed a real-time target update to overcome CT-to-body divergence. For nodules with no ideal bronchial access, the post-puncture scan confirmed the location of the needle tip within the lesion before creation of the tunnel, thereby ensuring the tunnel was established in the correct direction and avoiding unnecessary parenchymal injury and bleeding. The pre-freeze scan obtained after insertion of the cryoprobe confirmed that the cryoprobe was positioned within the target, ensuring that the biopsy would sample the intended tissue. The final scan on completion of the procedure was performed to check for pneumothorax and bleeding. While we recognize that two-dimensional fluoroscopy could be used for this purpose, we chose mobile CBCT because it also allowed visualization of post-biopsy cavities in some cases, providing additional confirmation that cryobiopsies had sampled the target tissue accurately if the cavities were located within the target lesion. Future studies should explore strategies to reduce radiation exposure while

maintaining accuracy and safety.

This study has some limitations. First, this was a retrospective study with limited sample size. Further studies are still required to validate the diagnostic performance and safety of the study procedure, and to compare different guidance methods and biopsy tools through multicenter randomized controlled trials. Second, this was a single-center study. All cases were performed by two experienced robotic bronchoscopists. The wide applicability of the study procedure remains to be demonstrated. Third, the proportion of patients with surgically confirmed diagnoses was low, as a substantial number of patients were unsuitable for or declined surgery, and the follow-up period was relatively short. While these factors may introduce potential false positives and false negatives, all negative diagnoses were classified as false negatives in this study. Thus, the false-negative rate is not underestimated. Fourth, although the study procedure had a high diagnostic yield, the improvement of the diagnostic performance compared to conventional guided bronchoscopy techniques and sampling tools needs to be further investigated by comparative studies.

Conclusions

In conclusion, ssRAB integrated with mobile CBCT-guided cryobiopsy had a high diagnostic yield for GGO-predominant PPNs with a favorable safety profile. Further studies are needed to validate its diagnostic performance and safety.

List of abbreviations

AAH, atypical adenomatous hyperplasia; CBCT, cone-beam computed tomography; CT, computed tomography; ENB, electromagnetic navigational bronchoscopy; GGO, ground-glass opacity; IQR, interquartile range; PD-L1, programmed death-ligand 1; PPN, peripheral pulmonary nodule; rEBUS, radial endobronchial ultrasound; SD, standard deviation; ssRAB, shape-sensing robotic-assisted bronchoscopy; VBN, virtual bronchoscopic navigation.

Declarations**Ethics approval and consent to participate**

This retrospective study was approved by the Ethics Committee of Shanghai Chest Hospital (IS25249) and the requirement for informed consent was waived.

Consent for publication

Not applicable. All data were anonymous.

Availability of data and materials

The datasets used and/or analysed during the current study are available from the corresponding author on reasonable request.

Competing interests

The authors declare that they have no competing interests.

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Authors' contributions

C. Z.: conceptualization, methodology, formal analysis, writing-original draft. F. X.: conceptualization, methodology, investigation, writing-review and editing. H. X.: investigation, data curation, writing-review. S. L.: investigation, writing-review. L. Y.: investigation, writing-review. D. X.: data curation, writing-review. J. S.: conceptualization, supervision, investigation, funding acquisition, project administration, writing-review and editing.

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